

Ideas

Could use the **CO2 emissions of electricity generation in Spain** to see the impact of ifca cloud and then other nations could do something similar

This research aims to estimate and make a projection of the yearly electricity and water consumption incurred by the current operating and planned computing centers and data transmission in Europe until 2030 → maybe the results of this paper can be used to see how things would improve if certain techniques were used

Moreover, the water impact from the energy component is directly influenced by the composition of the energy system. Therefore, the overall water impact of data usage can be dramatically reduced through transitioning the energy system towards renewables → or using energy-aware techniques

One possible application of this knowledge is for decision-making on where the next data centers should be deployed (logically, locations with the shortest distance to large population centers but in areas of low water stress) → maybe i can also map where the data centers are deployed currently and how many are in areas of high water stress (for a fairness analysis)

Wattnet facilitated informed decision making for workload scheduling and energy-aware operation of DCs while also enhancing transparency regarding the environmental impacts of electricity consumption for end users and policymakers. -> but they don't actually investigate this

On-site water consumption can be monitored and reported but this is still not a widespread practice → my work can be a framework to do this

Carbon emission and water consumption are represented as factors. And there are 2 branches one considers only the direct impact, and the other looks at the whole life cycle of the consumed energy.

Build framework that can be used by other DC to see their use and how that may impact other communities. How much energy produced by IFCA cloud and use the global/local WF and CF of Spain

Calculate PUE, WUE and WUE source (to show difference)